

Numerov Method for Bound States in the One-Dimensional Schrödinger Equation

Your Name

1 Introduction

This project studies the one-dimensional time-independent Schrödinger equation using the Numerov method combined with a parity-based shooting procedure. The goals are to validate the method on systems with known analytic solutions and then apply it to more interesting potentials, including a double well and a finite square well.

2 Computational approach

In dimensionless units, the Schrödinger equation is written as

$$\psi''(x) = 2[V(x) - E]\psi(x).$$

The wavefunction is integrated with the Numerov scheme on a uniform grid. For symmetric potentials, even and odd states are treated separately using parity conditions at $x = 0$. Eigenvalues are found by scanning for sign changes in the boundary mismatch and then refining them with bisection.

3 Results

Validation cases include the infinite square well and harmonic oscillator, where numerical energies are compared against exact results. Convergence is studied by varying both grid spacing and box size. For the quartic double well, the lowest even and odd states form nearly degenerate pairs; the splitting is further analyzed as a function of the barrier parameter. An additional finite square well is included to demonstrate the flexibility of the solver.

4 Discussion

The correctness of the code is verified by analytic benchmarks, normalization checks, and physical sanity tests such as the positivity of the double-well splitting. Convergence studies provide direct evidence that the numerical results are stable with respect to discretization choices. The wavefunction plots also show the expected parity and node structure.

5 Conclusion

The Numerov method provides an efficient and accurate tool for one-dimensional bound-state problems. In addition to reproducing known spectra, it captures physically interesting effects such as tunneling-induced energy splitting in a double well.